

NNIT CRM – API Integration Documentation

Production API Documentation for External System & n8n Automation Integration

CRM: NNIT CRM (NNIT Smart Parking & CRM)

Production Base URL: <https://chronolms.com>

Authentication Method: JWT (JSON Web Token via Bearer Header)

1. Integration Overview

NNIT CRM exposes RESTful APIs for external systems (such as **n8n**, custom portals, and automated workflows) to securely access CRM data. External systems must communicate with NNIT CRM through HTTPS APIs only. Direct database or server access is strictly prohibited.

```
External System / n8n Workflow
      |      HTTPS REST API (JSON)
      v
NNIT CRM Production API (https://chronolms.com)
      |      Django REST Framework + SimpleJWT
      v
NNIT CRM Database (PostgreSQL / MySQL)
```

2. Production API Base URL & API Groups

Base URL: <https://chronolms.com>

Active API Endpoint Groups:

```
https://chronolms.com/token/ (JWT Authentication)
https://chronolms.com/lead/lead/ (Leads Management)
https://chronolms.com/lead/customer/ (Customer Management)
https://chronolms.com/lead/lead-followups/ (Follow-up Records)
https://chronolms.com/lead/lead-faqs/ (Qualifying FAQ Master)
https://chronolms.com/api/quotation/ (Quotations)
https://chronolms.com/product/ (Products Catalog)
https://chronolms.com/api/amc/ (AMC Contracts & Visits)
```

3. API Interactive Documentation

Live Swagger UI and ReDoc OpenAPI documentations are available for interactive endpoint testing:

- **Swagger UI:** <https://chronolms.com/swagger/>
- **ReDoc Interface:** <https://chronolms.com/redoc/>

4. API Authentication (JWT)

NNIT CRM uses JWT (JSON Web Token) authentication. Static API keys are not used. All requests to protected endpoints must pass the access token in the standard HTTP Authorization header:

```
Authorization: Bearer <access_token>
```

5. Obtain Access Token

Send a POST request with valid service account credentials to generate an access token and refresh token.

```
POST https://chronolms.com/token/
```

```
{ "username": "SERVICE_ACCOUNT_USERNAME", "password": "SERVICE_ACCOUNT_PASSWORD" }
```

```
{ "refresh": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXLTJ5IiwiaWF0IjoiMTY0MjY0MjY0IiwiaXNjaWkiOiJ1b3RlciJ9", "access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXLTJ5IiwiaWF0IjoiMTY0MjY0MjY0IiwiaXNjaWkiOiJ1b3RlciJ9" }
```

6. Refresh Access Token

When an access token expires (default 60 minutes), send the refresh token to receive a new access token.

POST <https://chronolms.com/token/refresh/>

```
Request: { "refresh": "<refresh_token>" }
```

Response: { "access": "<new_access_token>" }

7. Verify Token

Verify if an existing access token is active and valid.

POST https://chronolms.com/token/verify/

Request: { "token": "<access_token>" } ==> Returns HTTP 200 OK if valid.

8. Dedicated Service Account for n8n

For n8n workflows and team integrations, create a dedicated service account in NNIT CRM rather than using employee credentials.

Recommended Account: Username: n8n_automation | Email: n8n.automation@nnitcrm.com | Role: API / Integration Service Account

9. Recommended System Permissions

Module	Endpoint Path	Recommended Access
Lead Management	/lead/lead/	Read, Create, Update, Filter, Search
Customer Management	/lead/customer/	Read, Create, Update, Lookup
Lead Follow-ups	/lead/lead-followups/	Read, Create, Update
Lead FAQ Master	/lead/lead-faqs/	Read Only
Quotations	/api/quotation/	Read / Create (If required)
AMC Contracts	/api/amc/	Read Only (If required)
User Management	/auth/staff/	No Access
System Settings	/admin/	No Access

10. Lead API Documentation

Endpoint: <https://chronolms.com/lead/lead/>

Supported HTTP Methods: GET, POST, PUT, PATCH, DELETE

10.1 Create Lead Request Payload & Fields

When creating a lead, pass the existing customer ID under the customer field. If creating via n8n, ensure the customer exists first (or create one using the Customer API).

Field	Type	Description / Allowed Values
customer	Integer	Required. Primary key of Customer record

requirements_details	Text	Requirements & specifications (e.g. 4 level stacker parking)
lead_source	String	Required. Enum: google_ads, indiamart, bni, justdial, reference, architect/interior_designer, builder, existing_customer, ka_staff, other
lead_source_input	JSON	Additional source context metadata object
status	String	Enum: open, in_process, close_win, close_loss, closed (Default: open)
lead_type	String	Type of lead (e.g. Commercial, Residential)
is_service_lead	String	Enum: sales, service, both
service_type	JSON	Service categories/types details
company_name	String	Company or site title
project_name	String	Name of project/site location
project_address	String	Address of project site
contact_person_name	String	Specific contact person for this lead
contact_person_number	String	Phone number of contact person
assign_to	Integer	User ID of CRM staff member assigned
enquiry_date	Date	Format: YYYY-MM-DD
followup_date	Date	Next follow-up date (Format: YYYY-MM-DD)
remarks	Text	Initial remarks or notes
is_qualified	Boolean	Lead qualification status (default: false)
qualifying_answers	JSON	Object mapping FAQ IDs to answers: {"1": "Budget 10L"}

Sample Create Lead Request (POST /lead/lead/):

```
{ "customer": 105, "requirements_details": "Customer requires 6-level puzzle parking system for commercial building.", "lead_source": "google_ads", "status": "open", "lead_type": "Commercial Parking", "is_service_lead": "sales", "company_name": "Apex Commercial Hub", "project_name": "Apex Tower Project", "project_address": "Plot 42, Vashi, Navi Mumbai", "contact_person_name": "Rajesh Malhotra", "contact_person_number": "9876543210", "enquiry_date": "2026-09-10", "followup_date": "2026-09-15", "remarks": "High priority lead from Google Ads campaign." }
```

11. Lead Status & Source Enums

Supported Lead Status Values: open | in_process | close_win | close_loss | closed

Supported Lead Source Values: google_ads | indiamart | bni | justdial | reference | architect/interior_designer | builder | existing_customer | ka_staff | other

12. Search, Filter, & Sort Leads

Search Endpoint: GET /lead/lead/?search=9876543210 (Searches Customer ID, Name, Contact, Email, Project Name, Lead Source)

Filtering Options:

```
GET /lead/lead/?status=open
GET /lead/lead/?assign_to=5
GET /lead/lead/?followup_date=2026-09-15
GET /lead/lead/?followup_today=true
GET /lead/lead/?overdue=true
GET /lead/lead/?date_from=2026-09-01&date_to=2026-09-30
GET /lead/lead/?followup_date_from=2026-09-10&followup_date_to=2026-09-20
```

Sorting Options: GET /lead/lead/?ordering=-date (Supported fields: date, followup_date, customer__name, status)

13. Fetch Latest Lead by Mobile Number

Fetch the latest lead associated with a given phone number (ideal for WhatsApp / IVR / n8n incoming call routing).

```
GET https://chronolms.com/lead/lead/latest-lead-by-mobile/?mobile=9876543210
```

14. Convert Lead to Customer

Converts a lead into a verified customer record, sets is_lead_only=False, marks is_converted=True, and updates lead status to close_win.

```
POST https://chronolms.com/lead/lead/{id}/convert-to-customer/
```

20. Customer API Documentation

Endpoint Base Path: https://chronolms.com/lead/customer/

Supported Operations: GET (list/retrieve), POST (create), PUT/PATCH (update), DELETE

Special Endpoints:

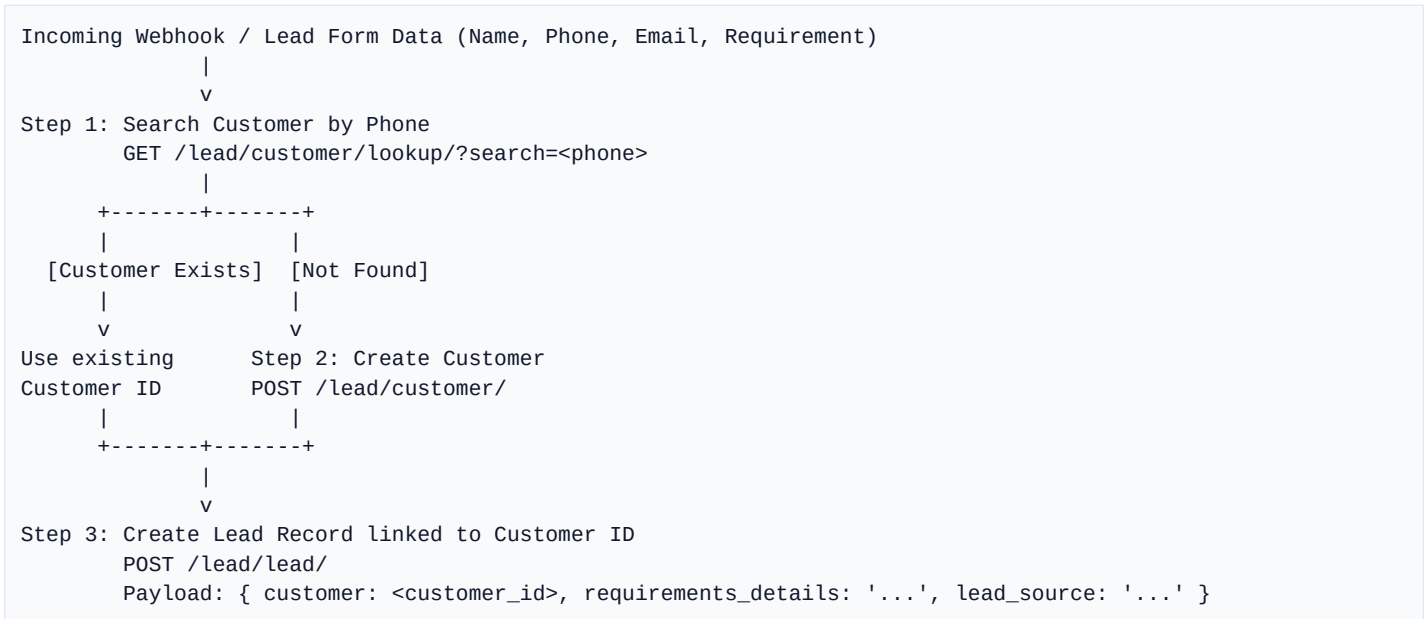
- GET /lead/customer/lookup/?search=9876543210 — Lookup customer by name, contact, or email across all records
- GET /lead/customer/{id}/leads/ — Fetch all leads belonging to this customer
- GET /lead/customer/{id}/followup-history/ — Fetch complete follow-up history for customer

Customer Object Fields Table:

Field Name	Type	Description
id	Integer	Read-only. Primary Key
name	String	Customer / Client Name (Required)
contact_number	String	Unique. Main contact phone number
secondary_contact_number	String	Alternative phone number
email	Email	Customer primary email address
secondary_email	Email	Secondary email address
poc_name	String	Point of contact person name
poc_contact_number	String	Point of contact phone number
land_line_no	String	Landline office phone
address	Text	Billing / Office address
city	String	Billing city
state	String	Billing state
pin_code	String	Billing PIN code
both_address_is_same	Boolean	If site address is same as billing (Default: false)
site_address	Text	Project site address
site_city	String	Project site city
site_state	String	Project site state
site_pin_code	String	Project site PIN code
gst	String	GST Identification Number (15 chars)
pan	String	PAN Number (10 chars)
is_lead_only	Boolean	Read-only. True if created from lead prior to conversion

21. Recommended Customer & Lead Creation Flow for n8n

Because contact_number on Customer is unique, external systems & n8n workflows MUST follow this standard lookup-first architecture to prevent duplicate entries:



22. Lead Follow-up API Documentation

Endpoint Base Path: <https://chronolms.com/lead/lead-followups/>

Supported Operations: GET, POST, PUT, PATCH, DELETE

Filter Follow-ups by Lead ID: GET /lead/lead-followups/?lead=105

Lead Follow-up Timeline: GET /lead/lead-followups/timeline/{lead_id}/

Lead Follow-up Fields Table:

Field Name	Type	Description & Enums
lead	Integer	Required. Primary key of Lead record
followup_date	Date	Required. Date of follow-up (YYYY-MM-DD)
followup_time	Time	Time of follow-up (HH:MM:SS)
next_followup_date	Date	Next scheduled follow-up date
remarks	Text	Follow-up remarks
discussion_notes	Text	Detailed discussion notes
interaction_type	String	Enum: call, whatsapp, email, video_call, in_person, demo, site_visit
client_response	String	Enum: very_positive, positive, neutral, negative, no_response, call_back_later
followup_status	String	Enum: completed, pending, scheduled (Default: completed)
status	String	Lead status update (open, in_process, close_win, close_loss, closed)
conducted_by	String	Team member who conducted follow-up
contacted_person	String	Name of person contacted
followup_summary	Text	Key summary points
client_commitment	Text	Client promises / action items
our_commitment	Text	Our promised deliverables / quote date
qualifying_info	JSON	Object containing site details, budget, decision maker, timeline
requirement_info	JSON	Object containing length, width, height, parking type, automation
faq_answers	Array	List of FAQ answer objects: [{ faq: 1, answer: 'Budget is 15L' }]

Sample Create Follow-up Request (POST /lead/lead-followups/):

```
{ "lead": 105, "followup_date": "2026-09-12", "followup_time": "14:30:00", "next_followup_date":
"2026-09-18", "interaction_type": "call", "client_response": "positive", "followup_status": "completed",
"status": "in_process", "remarks": "Discussed parking layout options with client.", "discussion_notes":
"Client requested proposal for 6-level puzzle parking.", "client_commitment": "Will share site CAD drawing
by Monday.", "our_commitment": "Will provide formal quotation within 48 hours.", "qualifying_info": {
"decision_maker": "Mr. Rajesh Malhotra (MD)", "budget_range": "25 - 30 Lakhs", "timeline": "Immediate
(within 1 Month)", "car_type": "suv" }, "faq_answers": [ { "faq": 1, "answer": "Budget confirmed around 28
Lakhs." }, { "faq": 2, "answer": "Decision maker is the Managing Director." } ] }
```

23. Lead FAQ Master API

Endpoint Base Path: <https://chronolms.com/lead/lead-faqs/>

Fields: id (Integer), question (String), is_active (Boolean), sort_order (Integer)

24. n8n Node Configuration Quick Reference Guide

When configuring an **n8n HTTP Request Node** to communicate with NNIT CRM:

- 1. Authentication:** Use Header Auth or automated JWT login sub-workflow. Set Header Key: Authorization and Value: Bearer `{{ $node['JWT Login'].json['access'] }}`
- 2. Content-Type:** Set Content-Type: application/json
- 3. Customer Lookup Node:** Method: GET | URL: [https://chronolms.com/lead/customer/lookup/?search={{ \\$json.phone }}](https://chronolms.com/lead/customer/lookup/?search={{ $json.phone }})
- 4. If Node:** Check if `{{ $json.length }}` > 0. If yes, take customer ID `{{ $json[0].id }}`. If no, route to Create Customer node.
- 5. Create Lead Node:** Method: POST | URL: <https://chronolms.com/lead/lead/>

25. Security Requirements & Production Checklist

- Do not share production server SSH keys, Django admin credentials, or database passwords in code repositories or n8n exported workflows.
- Always use HTTPS for API endpoints.
- Store the service account password securely in n8n Credentials or Environment Variables.

26. Final Integration & Handover Summary

Requirement	Production Status / Value
Production Base URL	https://chronolms.com
JWT Authentication Endpoint	POST https://chronolms.com/token/
JWT Refresh Endpoint	POST https://chronolms.com/token/refresh/
Lead API Endpoint	https://chronolms.com/lead/lead/
Customer API Endpoint	https://chronolms.com/lead/customer/
Lead Follow-up Endpoint	https://chronolms.com/lead/lead-followups/
Lead FAQ Master Endpoint	https://chronolms.com/lead/lead-faqs/
Swagger UI Documentation	https://chronolms.com/swagger/
ReDoc Documentation	https://chronolms.com/redoc/
n8n Dedicated Service Account	Username: n8n_automation (To be created in CRM)
Service Account Permissions	Lead, Customer, Follow-up (Read, Create, Update)